

Aarogyam / MedAdhere — IoT Smart Pill Dispenser

Complete Technical Specification, Firmware API Protocol & Hardware Architecture

FIRMWARE V3.2.0

HARDWARE V4.0

DJANGO CHANNELS WEBSOCKET

1. System & Hardware Architecture

The **Aarogyam / MedAdhere IoT Smart Pill Dispenser** is an autonomous, sensor-verified circular 4-compartment medication dispenser powered by an ESP32 microcontroller (or Arduino-compatible architecture). It operates in both **fully autonomous offline mode** (using an onboard DS3231 RTC and local NVS flash memory) and **cloud-synchronized online mode** via HTTP REST and real-time WebSockets.

Hardware Component	Interface / Pins	Operational & Physical Function
Stepper Motor (28BYJ-48 + ULN2003)	GPIO 13, 14, 27, 26	Drives 360° circular carousel. 1024 half-steps correspond precisely to 90° rotation per compartment.
Servo Lid Gate (SG90 / MG90S)	GPIO 25 (PWM)	Opens (90°) and closes (0°) the active compartment access aperture.
Ultrasonic Sensor (HC-SR04)	Trig: GPIO 19, Echo: GPIO 18	Detects patient hand proximity (≤ 15 cm) to trigger gate opening and record retrieval.
Precision Load Cell (1 kg + HX711)	DOUT: GPIO 35, SCK: GPIO 18	Reads sub-gram carousel weight before and after lid closure to verify pill removal.
Real-Time Clock (DS3231)	I2C: SDA 21, SCL 22 (0x68)	High-precision temperature-compensated RTC for autonomous schedule tracking without WiFi.
OLED Display (SSD1306 128×64)	I2C: SDA 21, SCL 22 (0x3C)	Renders medicine name, dosage instructions, time slot label, and system status.
DFPlayer Mini MP3 Audio	UART2: TX 17, RX 16	Voice audio guidance in Hindi/English (dose reminder, take medicine, taken, missed, unlock).
Hall Effect Sensor	GPIO 33	Detects carousel magnet for automatic homing to Compartment 1 reference index.
Buzzer & Indicator LED	Buzzer: GPIO 32, LED: GPIO 2	Multi-frequency chimes and visual blinking indicator during active dose windows.

2. Authentication Protocol

All communications between the IoT device and the cloud backend are strictly authenticated:

- HTTP REST Endpoints:** Header `X-Device-Key: <DEVICE_API_KEY>` (Validated by `DeviceAPIKeyAuthentication`).
- WebSocket Channel:** Query parameter `?device_key=<DEVICE_API_KEY>` (Validated by `DeviceKeyAuthMiddleware`).

3. Core Device Firmware APIs (Directly Called by Device)

GET /api/v1/iot/devices/{device_id}/config/

Description: Fetches the master configuration bundle. The device caches this in NVS flash and drives scheduled alarms from the onboard DS3231 RTC autonomously.

```
// Response (200 OK)
{
  "status": "success",
  "data": {
    "schedule_version": 4,
    "device_id": "e214a30b-c919-4d23-b3f1-80557b756cdd",
    "timezone": "Asia/Kolkata",
    "server_time_local": "2026-09-16T08:00:00+05:30",
    "total_weight_grams": 142.50,
    "is_gate_locked": false,
    "policy": {
      "dose_window_minutes": 60, "catchup_window_minutes": 30,
      "max_gate_opens": 4, "weight_settle_ms": 3000, "hand_detect_cm": 15
    },
    "compartments": [
      {
        "compartment_number": 1,
        "time_slot": "morning_before", "time": "08:00", "enabled": true,
        "content_weight_grams": 35.50, "expected_dose_reduction_grams": 1.25,
        "display_text": "Paracetamol 500mg\n1 Tab (Before Food)",
        "voice_text": "Dawai lene ka waqt ho gaya hai. Kripya Paracetamol lijiye.",
        "medicines": [{"medicine_uuid": "3fa8...", "name": "Paracetamol", "pill_weight_grams": 0.625, "qty_per_
      }
    ]
  }
}
```

GET /api/v1/iot/sync/time/

Description: Returns accurate Unix epoch and UTC/ISO timestamps to synchronize and calibrate the DS3231 RTC register on boot and drift detection.

```
// Response (200 OK)
{
  "status": "success",
  "data": {
    "unix_timestamp": 1789504215,
    "iso_time": "2026-09-15T20:30:15.123456Z",
    "utc_offset_seconds": 0
  }
}
```

POST /api/v1/iot/heartbeat/

Description: Periodic health beacon (5–10 mins). Records battery percentage, sensor statuses, and checks schedule version reconciliation.

```
// Request Body
{
  "device_id": "e214a30b-c919-4d23-b3f1-80557b756cdd",
  "battery_level": 92, "firmware_version": "3.2.0-ino", "wifi_strength": -58,
  "stepper_status": "ok", "servo_status": "ok", "ultrasonic_status": "ok",
  "rtc_time": "2026-09-16T08:00:00", "schedule_version": 3
}

// Response (200 OK)
{
  "status": "success",
  "data": {
    "server_unix_time": 1789504200, "schedule_version": 4,
    "config_stale": true, "gate_locked": false, "rtc_drift_seconds": 1
  }
}
```

POST /api/v1/iot/events/

Description: Emits real-time hardware, dispensing, and adherence events. Idempotent per `event_uuid`.

```
// Request Body
{
  "event_uuid": "c56d0e92-3482-4f32-a5d6-d0a92e106df2",
  "event_type": "DOSE_TAKEN", "compartment_num": 1, "medicine": "Paracetamol",
  "dose": 1, "occurred_at": "2026-09-16T08:02:15+05:30", "device_id": "e214a30b-c919-4d23-b3f1-80557b756cdd"
}

// Response (200 OK / 201 Created)
{
  "status": "success",
  "data": {"status": "accepted", "adherence_event_id": "9a6e1189-9821-4f81-9b63-07bf653068e1"}
}
```

POST /api/v1/iot/events/batch/

Description: Flushes offline NVS flash event queue when WiFi reconnects. Preserves original RTC timestamps (`occurred_at`).

```
// Request Body
{
  "events": [
    {"event_uuid": "1b9d...", "event_type": "DOSE_STARTED", "compartment_num": 1, "occurred_at": "2026-09-16T08:02:15+05:30"},
    {"event_uuid": "550e...", "event_type": "DOSE_TAKEN", "compartment_num": 1, "occurred_at": "2026-09-16T08:02:15+05:30"}
  ]
}
```

POST /api/v1/iot/events/weight-reading/

Description: Transmits raw settled load cell measurement after lid closes. Backend verifies whether ΔW matches prescribed reduction.

```
// Request Body: {"compartment_number": 1, "weight_grams": 84.15, "session_uuid": "b0f7..."}
// Response (200 OK)
{
  "status": "success",
  "data": {
    "dose_status": "taken", "actual_reduction_grams": 1.25,
    "expected_reduction_grams": 1.25, "is_valid": true, "gate_locked": false
  }
}
```

POST /api/v1/iot/devices/{device_id}/fill/measure/

Description: Guided refill mode. Reports total carousel weight to calculate exact per-pill weight: $(total - previous) / pills$.

```
// Request Body: {"compartment_number": 1, "total_weight_grams": 145.8, "medicine_id": "3fa8..."}
// Response (200 OK)
{
  "status": "success",
  "data": {
    "compartment_number": 1, "derived_weight_grams": 25.40,
    "pill_weight_grams": 0.8467, "schedule_version": 5
  }
}
```

GET /api/v1/iot/devices/{device_id}/commands/

Description: HTTP polling fallback for remote caregiver commands (sorted with HIGH priority commands first).

```
// Response (200 OK)
{
  "status": "success",
  "data": {
    "commands": [
      {"command_id": "f5e4...", "type": "DISPENSE_NOW", "payload": {"compartment": 2, "priority": "HIGH"}}
    ]
  }
}
```

4. Real-Time WebSocket Channel

Connection URL: `ws://<BACKEND_HOST>:<PORT>/ws/iot/device/<DEVICE_ID>/?device_key=<DEVICE_API_KEY>`

Direction	Frame Type	Payload Schema
Device → Server	ping	<code>{"type": "ping"}</code> (Sent every 30 seconds)
Server → Device	pong	<code>{"type": "pong", "server_unix_time": 1789504220}</code>
Server → Device	command	<code>{"type": "command", "command_id": "...", "command_type": "TRIGGER_DOSE", "payload": {"compartment": 1}}</code>
Device → Server	ack	<code>{"type": "ack", "command_id": "..."} (Acknowledges execution)</code>

5. Event Catalog & Hardware Action Table

Event Name	Trigger Condition	Hardware Reaction & Backend Action
DEVICE_BOOT	ESP32 power up or reset	Performs sensor self-test, homes carousel, requests daily schedule bundle.
DOSE_STARTED	RTC slot time match (IST)	Rotates stepper to target slot, tare reads load cell baseline, rings buzzer.
HAND_DETECTED	Ultrasonic sensor $\leq$ 15 cm	Opens servo lid (90°), displays prompt on OLED, plays audio track.
LID_CLOSED	Hand withdrawn from slot	Closes servo lid (0°), waits 3s weight stabilization, triggers scale reading.
WEIGHT_READING	Post-dose load cell read	Backend validates ΔW against expected reduction; marks session taken/partial.
DOSE_TAKEN	Intake verified	Decrements pill count in DB inventory, plays confirmation chime, notifies caregiver.
DOSE_MISSED	60-min window elapsed	Locks gate, triggers caregiver escalation notifications via Twilio & FCM.
LOW_BATTERY	Battery level < 15%	Logs critical power telemetry, alerts caregiver to plug in USB-C charger.

6. Remote Commands Catalog

Command Name	Parameters	Physical Hardware Execution
DISPENSE_NOW / TRIGGER_DOSE	<code>{"compartment": 1}</code>	Immediately rotates stepper to compartment, plays voice prompt, and dispenses medicine.
START_FILL_MODE	<code>{"compartment": 1, "medication_name": "Aspirin"}</code>	Rotates to slot 1, opens lid, displays medicine name on OLED, prepares scale for refill.
NEXT_COMPARTMENT	<code>{"compartment": 2, "medication_name": "Vitamin C"}</code>	Closes lid, pauses 500ms, rotates stepper to slot 2, and re-opens lid.
END_FILL_MODE	<code>{}</code>	Closes lid, chimes 3 times, locks in expected baseline weights, exits fill mode.
READ_FILL_WEIGHT	<code>{"medicine_id": "...", "compartment_number": 1}</code>	Settles scale for 3 seconds, reads HX711 grams, and posts reading to backend.
GATE_LOCK	<code>{}</code>	Closes lid immediately, sets locked flag, plays missed dose audio.
GATE_UNLOCK	<code>{}</code>	Clears locked flag, plays unlock audio, enables normal access.
SYNC_CONFIG	<code>{}</code>	Re-fetches schedule bundle from cloud backend and refreshes slot times.

7. Companion Caregiver & Mobile App APIs (JWT Authenticated)

Method	Endpoint	Purpose
POST	/api/v1/iot/devices/validate-code/	Validates device unique sticker registration code.
POST	/api/v1/iot/devices/link/	Links provisioned hardware dispenser to caregiver/patient account.
POST	/api/v1/iot/devices/<pk>/dispenser/setup/	Initializes 4 physical compartments with slot alarm times.
PATCH	/api/v1/iot/devices/<pk>/dispenser/compartments/<num>/time/	Updates scheduled time (HH:MM in IST) for any compartment.
POST	/api/v1/iot/devices/<pk>/dispenser/compartments/<num>/medicine/add/	Registers prescribed medicine to a compartment.
POST	/api/v1/iot/devices/<pk>/dispenser/fill/complete/	Confirms refill, locks in baseline load cell weights.
POST	/api/v1/iot/devices/<pk>/commands/queue/	Queues arbitrary commands (e.g. <code>DISPENSE_NOW</code> , <code>GATE_LOCK</code>).
POST	/api/v1/iot/dose/caregiver-unlock/	Remotely unlocks dispenser after missed dose or security lockout.
GET	/api/v1/iot/dose/history/?device_id=<uuid>	Retrieves historical dose adherence telemetry.
GET	/api/v1/iot/dose/alerts/	Lists recent missed doses, tampering, and lockout alerts.

8. End-to-End Dispensing Data Flow

[Device Boot / RTC Sync] Device → GET /api/v1/iot/sync/time/ → RTC DS3231 Calibrated Device → GET /api/v1/iot/devices/{id}/config/ → Schedule Bundle Saved in Flash Device ↔ ws://...ws/iot/device/{id}/ → Bi-Directional Command Channel Active [Autonomous Scheduled Dispensing Cycle] 1. DS3231 RTC hits Slot Time (e.g. 08:00 IST) 2. Device → POST /api/v1/iot/events/ (DOSE_STARTED) 3. Stepper rotates to Compartment 1 (1024 half-steps) 4. HX711 reads carousel baseline weight tare 5. DFPlayer plays Audio Prompt 1 & Buzzer chimes 6. Patient hand approaches (HC-SR04 ≤ 15 cm) → Servo Gate opens (90°) 7. Device → POST /api/v1/iot/events/ (LID_OPENED) 8. Hand withdraws → Servo Gate closes (0°) → POST (LID_CLOSED) 9. Scale settles (3s) → HX711 reads after-dose weight 10. Device → POST /api/v1/iot/events/weight-reading/ 11. Backend verifies ΔW → replies dose_status: "taken" 12. Device plays Confirmation Track 3; Inventory decremented in database